

TRANSITIVE PREFERENCE DECOMPOSITIONS: A HIGH-DIMENSIONAL PHASE TRANSITION

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ABSTRACT. In the alignment of large language models, the standard assumption that aggregate human preferences can be represented by a single reward model is fundamentally challenged by the Condorcet paradox. To resolve cyclic aggregate preferences, a natural approach is to partition a population of m voters into the minimum number of subgroups, $Z_{m,n}$, such that the aggregate preference over n alternatives is perfectly transitive within each subgroup. In this paper, we study the typical behavior of $Z_{m,n}$ under the impartial culture model in the high-dimensional asymptotic regime where $m, n \rightarrow \infty$. We establish an order-sharp phase transition at the critical scaling $m \asymp n^{5/3}$. Specifically, we show that with probability exponentially close to one, the population completely shatters ($Z_{m,n} = m$) when $m \lesssim n^{5/3}$. Conversely, when the crowd is sufficiently large ($m \gtrsim n^{5/3}$), statistical concentration takes over, and a simple two-group mixture is completely sufficient ($Z_{m,n} \leq 2$). This dichotomy reveals fundamental structural limits on the capacity of reward models in high-dimensional preference aggregation.

1. INTRODUCTION

Suppose m voters rank n alternatives denoted by $[n] := \{1, \dots, n\}$. Each individual has a linear preference ranking $\sigma \in \mathfrak{S}_n$, where $\mathfrak{S}_n$ is the set of all permutations on $[n]$. The aggregation of these individual preferences into a collective decision is a foundational problem in social choice theory, with applications spanning economics, political science, and computer science [SLB08, EK⁺10, Arr12, Sen17].

To formalize this aggregation in practice, one of the most fundamental and widely used mechanisms is the pairwise comparison. For any two distinct alternatives i and j , collect the aggregate preference $\mathcal{P}_{ij}$ which is defined as the fraction of voters who prefer i over j . Traditionally, social choice literature has focused on regimes where the number of alternatives n is small and the number of voters m is large. In these traditional scenarios, pairwise comparisons are heavily utilized because eliciting a full ranking of even a dozen alternatives is cognitively demanding and practically infeasible for human voters [Fis15, Mar14, Saa01]. Pairwise comparisons, by contrast, are easily elicited through simple surveys or head-to-head elections, forming the basis for Condorcet methods and tournament designs. In modern scenarios, the number of alternatives n can be exceptionally large. A prominent example is the alignment of large language models (LLMs) via reinforcement learning from human feedback (RLHF) [OWJ⁺22], where the alternatives represent all possible responses an LLM can generate given a prompt. In this regime, eliciting full rankings from human evaluators is fundamentally impossible, meaning the alignment process must rely exclusively on sampled pairwise comparison data.

The ultimate goal of aggregating these preferences is often to establish a universally accepted collective ranking of the alternatives. Under the pairwise comparison framework, this objective is achievable if the aggregate preference is *transitive*, meaning there exists a permutation $\sigma \in \mathfrak{S}_n$ such that $\mathcal{P}_{\sigma(i), \sigma(j)} > 1/2$ for every $i < j$. When this holds, σ serves as an unambiguous collective ranking. Standard alignment methods like RLHF intrinsically assume that such a transitive structure exists, as they attempt to represent aggregate human preferences using a single, static scalar reward model [XLX⁺25, LLS⁺25]. However, when there are more than two alternatives ($n \geq 3$), aggregate preferences arising from real-world data are notoriously prone to cycles, as demonstrated by the famous Condorcet paradox [dC⁺14, Geh06]. In the presence of such cycles, a consistent collective ranking simply does not exist. This fundamental contradiction casts significant doubt on the viability of reward maximization approaches. Recognizing this flaw has motivated recent paradigm shifts toward Nash learning from human feedback (NLHF) [MVC⁺24, SLL⁺25], which abandons the requirement for a global ranking and instead models preference alignment as a game-theoretic equilibrium.

Another natural resolution to the problem of cyclic aggregate preferences is to partition the m voters into subgroups, such that within each subgroup, the aggregate preference is perfectly transitive [CQY⁺24, XSL⁺25]. However, this raises a fundamental structural question: what is the optimal way to partition

the voters, and what is the minimal structural complexity required to eliminate all Condorcet cycles? If a population requires too many subgroups to achieve transitivity, the predictive power of any group-specific preference model is lost. In this paper, we are interested in the minimum number of such subgroups. Let $Z_{m,n}$ be the minimum number of voter groups into which the population can be partitioned such that, inside each group, the aggregate preference is transitive. [kz: np-hard?]

We analyze this problem within a probabilistic setting where all voters choose a random ranking uniformly and independently. This specific voting scheme is known as an *impartial culture* and serves as a standard baseline assumption in social choice theory [Gui52, GK68, Kel74, TRG03, Sau22]. An equivalent generative model for this scheme is to assign an independent random variable $U_i^k \sim \text{Unif}(0, 1)$ for every alternative $i \in [n]$ and every voter $k \in [m]$. Under this formulation, voter k prefers alternative i over j if and only if $U_i^k < U_j^k$.

To capture modern scenarios like LLM alignment, we must move beyond existing asymptotic frameworks. The classical social choice regime assumes a fixed number of alternatives n while the number of voters $m \rightarrow \infty$. Conversely, recent probabilistic treatments of the Condorcet paradox [Sau22, LLS+25] explore the opposite extreme, fixing the number of voters m while the alternative space $n \rightarrow \infty$ to demonstrate that the probability of admitting a transitive aggregate preference algebraically decays to zero. While theoretically foundational, holding either dimension constant fails to capture modern large-scale data paradigms fully, where the volume of sampled human feedback scales massively alongside the complexity of generative models.

Instead, we investigate the high-dimensional asymptotic regime, where both the number of voters m and the number of alternatives n grow to infinity simultaneously. In this setting, the structural complexity of the population is dictated not by the absolute sizes of m or n , but by their relative polynomial scaling. Our main results (Theorem 2.1 and Theorem 3.1) establish an order-sharp phase transition for the typical value of $Z_{m,n}$ under the impartial culture. Specifically, we demonstrate that the critical scaling threshold governing the partition size occurs at $m \asymp n^{5/3}$. A notable feature of this transition is that it is tight up to absolute constants and the results hold with probability exponentially close to 1. Informally, we have:

$$\begin{cases} \mathbb{P}(Z_{m,n} = 2) \geq 1 - \exp(-\Omega(m/n)) - \exp(-\Omega(n)) & \text{if } m \gtrsim n^{5/3}, \\ \mathbb{P}(Z_{m,n} = m) \geq 1 - \exp(-\Omega(n)) & \text{if } m \lesssim n^{5/3}. \end{cases}$$

[zk: use constant c .]

To interpret this phase transition, consider the balance between the complexity of the alternatives (n) and the wisdom of the crowd (m). When the number of alternatives is sufficiently large compared to the population size ($n \gtrsim m^{3/5}$), the sheer dimensionality of the choice space guarantees that virtually any subset of voters will inevitably disagree enough to form Condorcet cycles. Consequently, the population completely shatters: no two voters can be grouped together without breaking transitivity, forcing every individual into their own isolated subgroup ($Z_{m,n} = m$). Conversely, when the crowd is sufficiently large relative to the number of alternatives ($m \gtrsim n^{5/3}$), statistical concentration takes over. The noise of individual random preferences smooths out enough that the entire population can be cleanly bifurcated into just two rational sub-populations. Despite Condorcet paradox stated that cycles appear with high probability in the population as a whole, a simple two-group mixture model is completely sufficient to recover perfectly transitive preferences.

[kz: should say a little about proof innovation.]

While our results establish an order-sharp phase transition at the critical scaling $m \asymp n^{5/3}$, several open questions remain. A natural problem is to characterize the precise behavior of $Z_{m,n}$ in the critical regime. Investigating the size of each subgroup when $Z_{m,n} = 2$ is also an interesting direction. Finally, extending this analysis to non-uniform preference distributions beyond the impartial culture remains an important theoretical challenge.

The remainder of this paper is organized as follows. In Section 2, we establish the proof of Theorem 2.1. In Section 3, we establish the proof of Theorem 3.1.

1.1. Notation. Throughout this paper, we use c (possibly with subscripts or primes, e.g., c_1, c'_1) to denote strictly positive universal constants, whose value may vary from line to line. We also write $A \lesssim B$ to indicate $A \leq cB$, $B \gtrsim A$ for $A \lesssim B$, and $A \asymp B$ when $A \lesssim B$ and $B \lesssim A$. We also employ the standard asymptotic notations $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$. We denote the set of non-negative real numbers by $\mathbb{R}_+$. The standard inner product in $\mathbb{R}^d$ is denoted by $\langle \cdot, \cdot \rangle$.

For a nonempty voter set $S \subset [m]$, and any two distinct alternatives i and j , define the aggregate preference $\mathcal{P}_{ij}^S$ as the fraction of voters in S who prefer i over j . Define the strict-majority digraph G_S contains the directed edge $i \rightarrow j$ if and only if $\mathcal{P}_{ij}^S > 1/2$. A cycle in G_S is often called a *Condorcet cycle*.

For precision, we explicitly distinguish between acyclicity and transitivity of the aggregate preference. We say the aggregate preference $\mathcal{P}^S$ is *acyclic* if the corresponding strict-majority digraph G_S is acyclic. We say the aggregate preference $\mathcal{P}^S$ is *transitive* if there exists a permutation $\sigma \in \mathfrak{S}_n$ such that $\mathcal{P}_{\sigma(i), \sigma(j)}^S > 1/2$ for every $i < j$. It is straightforward to verify that $\mathcal{P}^S$ is acyclic if and only if there exists a permutation $\sigma \in \mathfrak{S}_n$ such that $\mathcal{P}_{\sigma(i), \sigma(j)}^S \geq 1/2$ for every $i < j$; consequently, transitivity implies acyclicity.

2. POPULATION FRAGMENTATION IN LARGE ALTERNATIVE SPACE

In this section, we analyze the regime where the complexity of the alternative space dominates the population size. Intuitively, when the number of alternatives n is sufficiently large, the sheer volume of pairwise comparison provides too many opportunities for cyclic conflicts to emerge. Consequently, if we attempt to group even a small number of voters together, the independent variations in their random preferences will almost inevitably manifest as a Condorcet cycle.

Theorem 2.1 formalizes this phenomenon. It establishes that when $n \gtrsim m^{3/5}$, the population completely shatters: with probability exponentially close to 1, no two voters can be placed in the same subgroup without violating transitivity. In other words, the only valid partition is the trivial one where every voter is isolated, yielding $Z_{m,n} = m$.

Theorem 2.1. *There are c_1, c_2 such that, for every $m \geq 3$ and $n \geq c_1 m^{3/5}$,*

$$\mathbb{P}(Z_{m,n} = m) \geq 1 - e^{-c_2 n}.$$

Now we begin the proof of Theorem 2.1. For the aggregate preference of a set of two voters to be transitive, the two voters must have identical rankings, which has a probability of $1/n!$. There are at most $\binom{m}{2}$ voter set of size two, and the union bound gives a probability of at most $\binom{m}{2}/n!$, which is smaller than e^{-cn} for sufficiently large n . Therefore, we only need to consider transitive voter sets of size no less than three, which must also be acyclic. For any $3 \leq s \leq m$, define

$$p_{s,n} := \mathbb{P}(G_{[s]} \text{ is acyclic}),$$

and let

$$N_s := \#\{S \subseteq [m] : |S| = s, G_S \text{ is acyclic}\}.$$

Markov's inequality and the union bound give

$$(1) \quad \mathbb{P}(\exists S \subseteq [m] : |S| \geq 3, G_S \text{ is acyclic}) \leq \sum_{s=3}^m \mathbb{P}(N_s \geq 1) \leq \sum_{s=3}^m \mathbb{E}N_s.$$

Every fixed s -subset has the law used in the definition of $p_{s,n}$, so

$$\mathbb{E}N_s = \binom{m}{s} p_{s,n},$$

and our goal is reduced to upper bounding $p_{s,n}$. By partitioning the n alternatives into independent blocks of size $\Theta(1)$, and noting that every block needs to be acyclic as well, we can obtain the following upper bound on $p_{s,n}$ decreasing exponentially in n :

Lemma 2.1 (Theorem 2.2 in [LLS⁺25]). *There exist c_1, c_2 such that*

$$p_{s,n} \leq c_1 e^{-c_2 n}$$

for $s \geq 3$ and $n \geq 3$.

However, this bound is not tight. Via central limit arguments and Gaussian-specific properties, it was shown in [Mos10, Proposition 1.6] that $p_{\infty,n} = e^{-\Theta(n^{5/3})}$, which surprisingly decays super-exponentially. The following theorem generalizes this statement to any $s \gtrsim n^{4/3}$. Its proof is provided in Section 2.1.

Theorem 2.2. *There exist c_1, c_2, c_3 such that, for every $n \geq c_1$ and $s \geq c_2 n^{4/3}$,*

$$p_{s,n} \leq e^{-c_3 n^{5/3}}.$$

Combining Lemma 2.1 and Theorem 2.2, we can obtain a better upper bound on $p_{s,n}$ for any $s \geq 3$. Specifically, for $s \lesssim n^{4/3}$, we can partition the n alternatives into blocks of size $\Theta(s^{3/4})$. There are $\Theta(n/s^{3/4})$ blocks in total. Applying Theorem 2.2 to each block and using the independence between blocks, we obtain the upper bound $e^{-\Theta(n\sqrt{s})}$ for $s \lesssim n^{4/3}$. The result is summarized in the following theorem:

Theorem 2.3. *There are c_1, c_2 such that, for every $n \geq c_1$ and $s \geq 3$,*

$$p_{s,n} \leq \exp\left[-c_2 n \min\{n^{2/3}, \sqrt{s}\}\right].$$

Proof. Take the constants c_1 and c_2 from Theorem 2.2. We can choose a sufficiently small constant $\eta > 0$ and a sufficiently large integer s_0 such that

$$b := \min\left\{n, \left\lfloor \eta s^{3/4} \right\rfloor\right\} \geq c_1$$

for every $s \geq s_0$, and such that $s \geq c_2 b^{4/3}$. Partition the alternatives into $q := \lfloor n/b \rfloor$ disjoint blocks of size b . By the independence between blocks and Theorem 2.2,

$$p_{s,n} \leq \left(e^{-c_2 b^{5/3}}\right)^q = e^{-c_2 q b^{5/3}}.$$

If $b = n$, then $qb^{5/3} = n^{5/3}$. If $b < n$, then $q \geq n/(2b)$ and, for s_0 large enough, $b \geq (\eta/2)s^{3/4}$. Hence

$$qb^{5/3} \geq \frac{n}{2} b^{2/3} \geq cn\sqrt{s}.$$

Thus the desired inequality holds for every $s \geq s_0$.

It remains to treat the finite set $3 \leq s < s_0$. By Lemma 2.1, $p_{s,n} \leq c_3 e^{-c_4 n}$ for n sufficiently large. Increasing c_1 if necessary, $n^{2/3} \geq \sqrt{s_0}$ for $n \geq c_1$, so

$$\min\{n^{2/3}, \sqrt{s}\} = \sqrt{s} \quad (3 \leq s < s_0).$$

Taking the minimum of $c_4/\sqrt{s}$ over this finite set of s and combining it with c_2 proves the theorem. $\square$

Using the bound from Theorem 2.3, we can immediately prove the remainder of Theorem 2.1. Without any loss of generality, we can assume $m > n^{4/3}$. For $3 \leq s \leq n^{4/3}$, Theorem 2.3 and the standard estimate $\binom{m}{s} \leq (em/s)^s$ give

$$\log \mathbb{E}N_s \leq s \log \frac{em}{s} - c_1 n \sqrt{s}.$$

Uniformly in this range,

$$\frac{s \log(em/s)}{n\sqrt{s}} \leq \frac{\sqrt{s} \log(em)}{n} \leq n^{-1/3} \log(em) = o(1).$$

Hence, for all sufficiently large n , $\mathbb{E}N_s \leq e^{-c'_1 n \sqrt{s}}$. Summing over this range, for sufficiently large n ,

$$\sum_{3 \leq s \leq n^{4/3}} \mathbb{E}N_s \leq n^{4/3} e^{-c'_1 n} \leq e^{-c'_1 n}.$$

For $s > n^{4/3}$, Theorem 2.3 gives $p_{s,n} \leq e^{-c_2 n^{5/3}}$. Therefore for sufficiently large n ,

$$\sum_{s > n^{4/3}} \mathbb{E}N_s \leq 2^m e^{-c_2 n^{5/3}} \leq e^{-c'_2 n^{5/3}}.$$

By (1), we thus have

$$\mathbb{P}(\exists S \subseteq [m] : |S| \geq 3, G_S \text{ is acyclic}) \leq e^{-cn}.$$

This excludes every transitive voter set of size at least three. As we have already excluded transitive voter set of size two, the proof is completed.

2.1. Proof of Theorem 2.2. In this section, we leverage an observation of [Mos10] and show that in order to prove Theorem 2.2, it is enough to prove Theorem 2.4, which is a statement about the random variables R_{ij} that is defined in (4) below.

For any pair of alternatives $i \neq j$, define the normalized aggregate preference

$$M_{ij} := \sqrt{s}(\mathcal{P}_{ij}^{[s]} - \mathcal{P}_{ji}^{[s]}).$$

Then acyclicity means that there exists a permutation $\sigma \in \mathfrak{S}_n$ such that $M_{\sigma(i), \sigma(j)} \geq 0$ for every $i < j$. Define the following functions

$$g(u) := 1 - 2u, \quad \psi(u) := g(u \bmod 1).$$

Then for every $u_i \neq u_j$, we have

$$(2) \quad \mathbf{1}\{u_i < u_j\} - \mathbf{1}\{u_j < u_i\} = g(u_i) - g(u_j) + \psi(u_j - u_i).$$

Consequently, for s voters, the normalized aggregate preference admits the following decomposition:

$$(3) \quad M_{ij} = A_i - A_j + R_{ij},$$

where

$$A_i := \frac{1}{\sqrt{s}} \sum_{k=1}^s g(U_i^k)$$

and

$$(4) \quad R_{ij} := \frac{1}{\sqrt{s}} \sum_{k=1}^s X_{ij}^k, \quad X_{ij}^k := \psi(U_j^k - U_i^k).$$

For the random variables A_i , as the summands $g(U_i^k)$ are independent, centered, and lie in $[-1, 1]$, Hoeffding's inequality gives the following lemma:

Lemma 2.2. *For every alternative $i \in [n]$ and $t > 0$,*

$$\mathbb{P}(|A_i| > t) \leq 2e^{-t^2/2}.$$

Moreover, the variables $A_1, \dots, A_n$ are independent.

Before stating the lemma that records the properties of the random variables $\{X_{ij}\}_{i \neq j}$, we briefly set up some graph-theoretic notation. We consider a simple graph $H = (\mathcal{V}, \mathcal{E})$ consisting of a vertex set $\mathcal{V} \subset [n]$ and an edge set $\mathcal{E}$. For any subset of edges $\mathcal{F} \subseteq \mathcal{E}$, we let $\mathcal{V}(\mathcal{F})$ denote the subset of vertices incident to at least one edge in $\mathcal{F}$. The pair $(\mathcal{V}(\mathcal{F}), \mathcal{F})$ then forms the *edge subgraph* induced by $\mathcal{F}$.

Lemma 2.3. *Let $H = (\mathcal{V}, \mathcal{E})$ be a simple graph. For one voter, define the random variables $(X_e)_{e \in \mathcal{E}}$ as in (4) with an arbitrary orientation on each edge. Then:*

- (i) *if $\mathcal{F} \subseteq \mathcal{E}$ is a forest, the variables $(X_e)_{e \in \mathcal{F}}$ are independent and each is $\text{Unif}[-1, 1]$;*
- (ii) *if measurable functions $a_e : [-1, 1] \rightarrow \mathbb{R}$ satisfy $\mathbb{E}a_e(X_e) = 0$ and the edge subgraph $(\mathcal{V}(\mathcal{F}), \mathcal{F})$ has a vertex of degree one, then*

$$\mathbb{E} \prod_{e \in \mathcal{F}} a_e(X_e) = 0.$$

Proof. For an oriented edge $e = (i, j)$, we have $X_e = \psi(U_j - U_i)$. If one endpoint, say U_i , is fixed, then $(U_j - U_i) \bmod 1$ is uniform on $[0, 1]$. Since $\psi(u) = 1 - 2(u \bmod 1)$, it follows that X_e is uniform on $[-1, 1]$.

We first prove part (i) by induction on $|\mathcal{F}|$. The claim is trivial when $\mathcal{F}$ is empty. Otherwise, since $\mathcal{F}$ is a forest, its edge subgraph has a leaf vertex v . Let e be the unique edge in $\mathcal{F}$ incident to v , and condition on all variables $(U_u)_{u \neq v}$. Regardless of the orientation of e , we can write $X_e = \psi(\pm U_v + c)$, where c is fixed under the conditioning. Hence the conditional law of X_e is $\text{Unif}[-1, 1]$ and does not depend on $(U_u)_{u \neq v}$. In particular, X_e is independent of $(X_f)_{f \in \mathcal{F} \setminus \{e\}}$. Removing v and e still leaves a forest, then the induction hypothesis completes the proof of (i).

For part (ii), let v be a degree-one vertex of the edge subgraph induced by $\mathcal{F}$, and let e be its unique incident edge. Condition again on $(U_u)_{u \neq v}$. All variables X_f with $f \in \mathcal{F} \setminus \{e\}$ are then fixed, while X_e remains uniform on $[-1, 1]$. Therefore,

$$\mathbb{E} \left[\prod_{f \in \mathcal{F}} a_f(X_f) \middle| (U_u)_{u \neq v} \right] = \left(\prod_{f \in \mathcal{F} \setminus \{e\}} a_f(X_f) \right) \mathbb{E}[a_e(X_e)] = 0.$$

Taking expectations then proves (ii). $\square$

For the random variables $\{R_e\}_{e \in \mathcal{E}}$, we have the following lemma, whose proof is detailed in Section 2.2.

Theorem 2.4. *There exist universal constants $c_1, c_2 > 0$ such that the following holds. Let $H := (\mathcal{V}, \mathcal{E})$ be a simple graph with maximum degree Δ , and construct the residual sums $(R_e)_{e \in \mathcal{E}}$ as in (4) with an arbitrary orientation on each edge. If*

$$s \geq c_1 \Delta^2,$$

then

$$\mathbb{P} \left(R_e \geq -\frac{1}{8} \quad \forall e \in \mathcal{E} \right) \leq \exp(-c_2 |\mathcal{E}|).$$

Given Theorem 2.4, we can adapt the idea in [Mos10, Section 7.5.3] to finish the proof of Theorem 2.2. Consider a truncation scale $T = \Theta(n^\alpha)$ and discard all alternatives with $|A_i| > T$. By Lemma 2.2 and a union bound, insisting that $\Theta(n)$ alternatives remain costs a probability of $2^n(2\exp(-T^2/2))^n = \exp(-\Theta(n^{1+2\alpha}))$. Partition $[-T, T]$ into bins of length $1/8$. Roughly speaking, each bin contain $d \approx \Theta(n/T) = \Theta(n^{1-\alpha})$ alternatives. Any two alternatives $i \neq j$ within the same bin satisfies $|A_i - A_j| \leq 1/8$ by definition, so that $R_{ij} \geq -1/8$ provided that the aggregate preference is acyclic ($M_{ij} \geq 0$). Construct the simple graph H using all such ij , then it has $|\mathcal{V}| = \Theta(n)$ vertices, degree $\Delta \approx d = \Theta(n^{1-\alpha})$ and $|\mathcal{E}| \approx |\mathcal{V}|\Delta = \Theta(n^{2-\alpha})$. Therefore by Theorem 2.4, insisting acyclicity costs a probability of $\exp(-\Theta(|\mathcal{E}|)) = \exp(-\Theta(n^{2-\alpha}))$. Choosing the optimal $\alpha = 1/3$ gives us the desired rate.

Proof of Theorem 2.2. Fix a proposed ranking $\sigma \in \mathfrak{S}_n$ and relabel the alternatives so that $\sigma = \text{Id}$. By the definition of acyclicity, validity of this ranking requires

$$(5) \quad M_{ij} \geq 0 \quad (1 \leq i < j \leq n).$$

Set $T := n^{1/3}$ and call an alternative *good* if $|A_i| \leq T$. By Lemma 2.2, the bad indicators are independent and each has probability at most $2e^{-T^2/2}$. Thus

$$(6) \quad \mathbb{P}(\text{fewer than } n/2 \text{ alternatives are good}) \leq 2^n \left(2e^{-T^2/2}\right)^{n/2} \leq e^{-c_0 n^{5/3}}$$

for all sufficiently large n .

Now work on the event that at least $n/2$ alternatives are good. Partition $[-T, T]$ into K half-open intervals of length at most $\delta := 1/8$, referred to as *bins*, where

$$K := \lceil 2T/\delta \rceil \asymp n^{1/3}.$$

A bin *assignment* records, for every alternative, either the index of the bin containing A_i or the symbol 0 if $|A_i| > T$. There are at most $(K+1)^n$ assignments.

Fix one assignment having at least $n/2$ nonzero entries. Put

$$d := \left\lfloor \frac{n}{8K} \right\rfloor \asymp n^{2/3}.$$

For each bin, list its assigned alternatives in increasing label order, divide the list into consecutive *blocks* of size d , and discard the last block if its size is smaller than d . Within each block, include every pair of alternatives as an edge. Let H be the union of these complete graphs. Then $\Delta_H = d - 1$. In each of the K bins fewer than d alternatives are discarded, so fewer than $dK \leq n/8$ good alternatives are discarded in total. Therefore, at least $3n/8$ vertices belong to complete d -blocks and

$$|\mathcal{E}_H| = \frac{d-1}{2} \times (\text{number of vertices in complete } d\text{-blocks}) \asymp n^{5/3}.$$

Every selected edge joins two alternatives in the same bin. It is oriented from the smaller label i to the larger label j , so $i < j$. On the event (5), the decomposition (3) gives

$$R_{ij} = M_{ij} - (A_i - A_j) \geq A_j - A_i \geq -\delta = -\frac{1}{8}.$$

The graph H is deterministic once the bin assignment is fixed. If c_2 in the statement of Theorem 2.2 is sufficiently large, then $s \geq c_1 \Delta_H^2$, and Theorem 2.4 yields

$$\mathbb{P}\left(R_e \geq -\frac{1}{8} \quad \forall e \in \mathcal{E}_H\right) \leq e^{-c|\mathcal{E}_H|} \leq e^{-cn^{5/3}}.$$

Summing over assignments,

$$(7) \quad \mathbb{P}(\text{the fixed order is valid and at least } n/2 \text{ are good}) \leq (K+1)^n e^{-cn^{5/3}} \leq e^{-c'n^{5/3}},$$

because $n \log(K+1) = O(n \log n) = o(n^{5/3})$.

Combining (6) and (7) bounds the probability that this one order is valid by $e^{-c''n^{5/3}}$. There are $n!$ possible orders, and $\log(n!) = O(n \log n) = o(n^{5/3})$. A final union bound proves the desired result. $\square$

2.2. Proof of Theorem 2.4. In this section, we reduce Theorem 2.4 to a decoupling inequality (Theorem 2.5), which bounds the joint expectation of the dependent random variables $\{R_e\}_{e \in \mathcal{E}}$ by their product of marginals, up to a multiplicative error factor.

First, we reduce the problem of bounding the probability in Theorem 2.4 to bounding the expectation of the dependent random variables $\{R_e\}_{e \in \mathcal{E}}$. Let

$$(8) \quad h(x) := \begin{cases} 1, & x \leq -\frac{1}{4}, \\ -8x - 1, & -\frac{1}{4} < x < -\frac{1}{8}, \\ 0, & x \geq -\frac{1}{8}, \end{cases} \quad f(x) := e^{-h(x)}.$$

Then $f : \mathbb{R} \rightarrow [e^{-1}, 1]$, $\log f$ is 8-Lipschitz, $f(x) = 1$ for $x \geq -1/8$. Therefore,

$$\mathbf{1} \left\{ R_e \geq -\frac{1}{8} \quad \forall e \in \mathcal{E} \right\} \leq \prod_{e \in \mathcal{E}} f(R_e).$$

Next, let $V_1, \dots, V_s$ be i.i.d. $\text{Unif}[-1, 1]$ and define

$$Y_s := \frac{1}{\sqrt{s}} \sum_{k=1}^s V_k.$$

Then Y_s has the same marginal as R_e . The next lemma records an upper bound of $\mathbb{E}f(Y_s)$, whose proof can be found in Section A.

Lemma 2.4. *There is a universal constant $0 < \rho_0 < 1$ such that for every $s \geq 1$, $\mathbb{E}f(Y_s) \leq \rho_0$.*

The following theorem quantifies the error of replacing R_e with its product marginals. Its proof is detailed in Section 2.3.

Theorem 2.5. *Let $H = (\mathcal{V}, \mathcal{E})$ be a simple graph with maximum degree Δ , and construct the residual sums $(R_e)_{e \in \mathcal{E}}$ as in (4) with an arbitrary orientation on each edge. Let $f : \mathbb{R} \rightarrow (0, \infty)$ satisfy*

$$(9) \quad |\log f(x) - \log f(y)| \leq L|x - y| \quad (x, y \in \mathbb{R}).$$

Put

$$q_s := e^{2L/\sqrt{s}} - 1.$$

If $8\Delta q_s < 1$, then

$$\mathbb{E} \prod_{e \in \mathcal{E}} f(R_e) \leq (\mathbb{E}f(Y_s))^{|\mathcal{E}|} \exp\left(\frac{s|\mathcal{E}|}{\Delta^2} \frac{(8\Delta q_s)^3}{1 - 8\Delta q_s}\right).$$

In particular, Theorem 2.5 implies that there are universal constants $c_0, c_1 > 0$ such that, whenever

$$s \geq c_0 L^2 \Delta^2,$$

we have

$$(10) \quad \mathbb{E} \prod_{e \in \mathcal{E}} f(R_e) \leq (\mathbb{E}f(Y_s))^{|\mathcal{E}|} \exp\left(c_1 L^3 \frac{|\mathcal{E}| \Delta}{\sqrt{s}}\right).$$

To prove this, first note that if $2L/\sqrt{s} \leq 1$, then $q_s \leq 4L/\sqrt{s}$. Therefore, if $s \geq c_0 L^2 \Delta^2$ with c_0 sufficiently large, the above estimate is valid and $8\Delta q_s \leq 32L\Delta/\sqrt{s} \leq 1/2$. Hence we have

$$\frac{s|\mathcal{E}|}{\Delta^2} \frac{(8\Delta q_s)^3}{1 - 8\Delta q_s} \leq c_1 L^3 \frac{|\mathcal{E}| \Delta}{\sqrt{s}},$$

which gives (10). Applying (10) with f defined in (8) and Lemma 2.4 implies, whenever $s \geq c_0 \cdot 64\Delta^2$,

$$\mathbb{P}\left(R_e \geq -\frac{1}{8} \quad \forall e \in \mathcal{E}\right) \leq \exp\left[-a_0 |\mathcal{E}| + c_1 8^3 \frac{|\mathcal{E}| \Delta}{\sqrt{s}}\right],$$

where $a_0 := -\log \rho_0 > 0$. Choose c_0 sufficiently large that the second exponent is at most $a_0 |\mathcal{E}|/2$. Therefore, Theorem 2.4 holds with $c_2 = a_0/2$.

2.3. Proof of Theorem 2.5. In this section, we reduce the proof of Theorem 2.5 to Lemma 2.5, a pure combinatorial statement that bounds the number of generalized cycles in a simple graph. For a simple graph $H = (\mathcal{V}, \mathcal{E})$, we call any connected nonempty edge set $\gamma \subseteq \mathcal{E}$ whose induced edge subgraph has minimum degree at least two a *generalized cycle*. Let $\mathcal{L}(H)$ be the collection of all generalized cycles.

Let $\tilde{X} = (\tilde{X}_e)_{e \in \mathcal{E}}$ have independent $\text{Unif}[-1, 1]$ coordinates. Fix any vector $w = (w_e)_{e \in \mathcal{E}}$ and define

$$m_e(w) := \mathbb{E} f\left(w_e + \frac{\tilde{X}_e}{\sqrt{s}}\right), \quad a_{e,w}(x) := \frac{f(w_e + x/\sqrt{s})}{m_e(w)} - 1.$$

Then $\mathbb{E} a_{e,w}(\tilde{X}_e) = 0$. Since the true coordinate X_e has the same $\text{Unif}[-1, 1]$ marginal law, also $\mathbb{E} a_{e,w}(X_e) = 0$. By (9), for $x, y \in [-1, 1]$,

$$\frac{f(w_e + x/\sqrt{s})}{f(w_e + y/\sqrt{s})} \leq e^{2L/\sqrt{s}} = q_s + 1.$$

Since $m_e(w)$ lies between the minimum and maximum of $f(w_e + x/\sqrt{s})$ over $x \in [-1, 1]$,

$$(11) \quad |a_{e,w}(x)| \leq q_s.$$

For one true voter residual vector X , expand

$$\mathbb{E}_X \prod_{e \in \mathcal{E}} f\left(w_e + \frac{X_e}{\sqrt{s}}\right) = \prod_{e \in \mathcal{E}} m_e(w) \sum_{\mathcal{F} \subseteq \mathcal{E}} \mathbb{E}_X \prod_{e \in \mathcal{F}} a_{e,w}(X_e).$$

By Lemma 2.3, a nonempty term vanishes whenever the edge subgraph $\mathcal{F}$ has a degree-one vertex. Every surviving $\mathcal{F}$ is therefore a vertex-disjoint union of generalized cycles. Using (11), taking absolute values term by term, and then dropping the vertex-disjointness restriction,

$$\begin{aligned} \sum_{\mathcal{F} \subseteq \mathcal{E}} \mathbb{E}_X \prod_{e \in \mathcal{F}} a_{e,w}(X_e) &\leq 1 + \sum_{\substack{\mathcal{F}: \text{vertex-disjoint union} \\ \text{of generalized cycles}}} q_s^{|\mathcal{F}|} \\ &\leq \prod_{\gamma \in \mathcal{L}(H)} (1 + q_s^{|\gamma|}) \\ &\leq \exp(\Lambda_H(q_s)), \end{aligned}$$

where $\Lambda_H(q) := \sum_{\gamma \in \mathcal{L}(H)} q^{|\gamma|}$. On the other hand, independence of the coordinates of $\tilde{X}$ gives

$$\mathbb{E}_{\tilde{X}} \prod_{e \in \mathcal{E}} f\left(w_e + \frac{\tilde{X}_e}{\sqrt{s}}\right) = \prod_{e \in \mathcal{E}} m_e(w).$$

Thus, pointwise in w ,

$$(12) \quad \mathbb{E}_X \prod_{e \in \mathcal{E}} f\left(w_e + \frac{X_e}{\sqrt{s}}\right) \leq e^{\Lambda_H(q_s)} \mathbb{E}_{\tilde{X}} \prod_{e \in \mathcal{E}} f\left(w_e + \frac{\tilde{X}_e}{\sqrt{s}}\right).$$

Now let $X^{(1)}, \dots, X^{(s)}$ be the true voter vectors and let $\tilde{X}^{(1)}, \dots, \tilde{X}^{(s)}$ be independent vectors having independent uniform coordinates. For $j = 0, \dots, s$, define

$$S^{(j)} := \frac{1}{\sqrt{s}} \left(\sum_{v=1}^j X^{(v)} + \sum_{v=j+1}^s \tilde{X}^{(v)} \right), \quad Z_j := \mathbb{E} \prod_{e \in \mathcal{E}} f(S_e^{(j)}).$$

Conditioning on all vectors except the j th and applying (12) yields

$$Z_j \leq e^{\Lambda_H(q_s)} Z_{j-1}.$$

Iteration gives $Z_s \leq e^{s\Lambda_H(q_s)} Z_0$. In $S^{(0)}$, all edge coordinates are independent and each has law Y_s , so

$$Z_0 = (\mathbb{E} f(Y_s))^{|\mathcal{E}|}.$$

Combining this with Lemma 2.5 below, which bounds the number of generalized cycles, proves Theorem 2.5.

Lemma 2.5. Let $H = (\mathcal{V}, \mathcal{E})$ be a simple graph with maximum degree $\Delta \geq 1$. For every $s \geq 3$,

$$\#\{\gamma \in \mathcal{L}(H) : |\gamma| = s\} \leq |\mathcal{E}| 8^s \Delta^{s-2}.$$

Consequently, whenever $8\Delta q < 1$,

$$\Lambda_H(q) := \sum_{\gamma \in \mathcal{L}(H)} q^{|\gamma|} \leq \frac{|\mathcal{E}|}{\Delta^2} \frac{(8\Delta q)^3}{1 - 8\Delta q}.$$

2.4. Proof of Lemma 2.5. For a simple graph $H = (\mathcal{V}, \mathcal{E})$ and a vertex subset $S \subseteq \mathcal{V}$, let $H[S]$ denote the *subgraph induced by S* , that is, the graph with vertex set S and edge set $\{\{u, v\} \in \mathcal{E} : u, v \in S\}$. The *line graph* $L(H) = (\mathcal{V}_L, \mathcal{E}_L)$ of H has vertex set $\mathcal{V}_L = \mathcal{E}$, where two vertices $e_1, e_2 \in \mathcal{V}_L$ are adjacent in $L(H)$ if and only if e_1 and e_2 share a common endpoint in H .

Lemma 2.6. *Let $H = (\mathcal{V}, \mathcal{E})$ be a simple graph of maximum degree at most Δ , let $A \subseteq \mathcal{V}$ be a fixed connected set of size a , and let $s \geq a$. The number of connected vertex sets S satisfying $A \subseteq S$ and $|S| = s$ is at most*

$$4^s \Delta^{s-a}.$$

The proof is a simple extension of the classical argument by Bollobás, see e.g. [Bol06, Chapter 3.45].

Proof. Fix an ordering of A and, at every vertex of H , an ordering of its at most Δ incident edges. Given a connected set $S \supseteq A$, start with all vertices of A marked as discovered and run a deterministic breadth-first search of the induced subgraph $H[S]$, using root and neighbor orders we have fixed. Every vertex of $S \setminus A$ receives a unique parent that was discovered earlier. The resulting parent edges form an ordered rooted forest with the a prescribed roots A and $s - a$ nonroot vertices.

The number of ordered rooted forest shapes with s total vertices and the a prescribed ordered roots is at most 4^s . Indeed, adjoin one super-root and make the a roots its ordered children. This gives an injective map into the set of plane rooted trees with $s + 1$ vertices. The latter set has Catalan cardinality $C_s = \frac{1}{s+1} \binom{2s}{s} \leq 4^s$. For each of the $s - a$ parent-child edges outside the prescribed root set, the child is determined from its parent by one of at most Δ incident-edge indices. The deterministic search makes this encoding injective. Therefore the number of possible S is at most $4^s \Delta^{s-a}$. $\square$

Now we are ready to prove Lemma 2.5.

Proof of Lemma 2.5. Every generalized cycle contains a simple cycle. Every such cycle has length $\ell \geq 3$. The number of simple cycles of length ℓ is at most

$$2|\mathcal{E}| \Delta^{\ell-2}.$$

Indeed, choose an oriented starting edge in at most $2|\mathcal{E}|$ ways and then choose each of the next $\ell - 2$ vertices in at most Δ ways. The closing condition is ignored, which only enlarges the count.

Fix a cycle C of length ℓ . Any connected edge set of size s containing C corresponds to a connected vertex set of size s in $L(H)$ containing the ℓ edges of C . Since H has maximum degree Δ , any edge $e = \{u, v\} \in \mathcal{E}$ is incident to at most $2(\Delta - 1) < 2\Delta$ other edges, so $L(H)$ has maximum degree at most 2Δ . Lemma 2.6 bounds the number of such supersets by

$$4^s (2\Delta)^{s-\ell}.$$

Summing over the possible cycle lengths and allowing multiple counting of the same generalized cycle,

$$\begin{aligned} \#\{\gamma \in \mathcal{L}(H) : |\gamma| = s\} &\leq \sum_{\ell=3}^s 2|\mathcal{E}| \Delta^{\ell-2} 4^s (2\Delta)^{s-\ell} \\ &= 2|\mathcal{E}| 4^s \Delta^{s-2} \sum_{\ell=3}^s 2^{s-\ell} \\ &\leq |\mathcal{E}| 8^s \Delta^{s-2}. \end{aligned}$$

Summing the resulting geometric series gives

$$\begin{aligned} \Lambda_H(q) &\leq |\mathcal{E}| \sum_{s=3}^{\infty} 8^s \Delta^{s-2} q^s \\ &= \frac{|\mathcal{E}|}{\Delta^2} \frac{(8\Delta q)^3}{1 - 8\Delta q}. \end{aligned}$$

$\square$

3. TWO-GROUP SUFFICIENCY VIA THE WISDOM OF THE CROWD

In contrast to the extreme fragmentation observed in the previous section, we now turn our attention to the regime where the voter population is sufficiently large relative to the complexity of the alternative space. In this setting, the “wisdom of the crowd” prevails. Through statistical concentration, the chaotic variations of individual random preferences smooth out, enabling us to cleanly partition the entire population into just two perfectly transitive subgroups.

Theorem 3.1 formalizes this sufficiency. It establishes that as long as $m \gtrsim n^{5/3}$, the probability of requiring more than two groups decays exponentially with respect to the ratio m/n .

Theorem 3.1. *There are c_1, c_2 such that, for every $n \geq 3$ and every $m \geq c_1 n^{5/3}$,*

$$\mathbb{P}(Z_{m,n} \leq 2) \geq 1 - \exp\left(-c_2 \frac{m}{n}\right).$$

Before proceeding to the proof, we note that while the theorem bounds the probability of $Z_{m,n} \leq 2$, the typical value in this regime is exactly 2. By Lemma 2.1, the probability that the entire undivided population is transitive ($Z_{m,n} = 1$) is exponentially small, specifically $\mathbb{P}(Z_{m,n} = 1) \lesssim e^{-cn}$. Combining this with Theorem 3.1, we immediately obtain that for every $m, n \geq 3$ with $m \geq c_1 n^{5/3}$,

$$\mathbb{P}(Z_{m,n} = 2) \geq 1 - \exp\left(-c_2 \frac{m}{n}\right) - e^{-c_3 n}.$$

This confirms that a two-group partition is both necessary and sufficient with high probability.

Now we begin the proof of Theorem 3.1. Let $\mathcal{E} := \{(i, j) : 1 \leq i < j \leq n\}$. For $e = (i, j) \in \mathcal{E}$ and voter $k \in [m]$, define the random variable

$$Y_e^k = Y_{ij}^k := \mathbf{1}\{U_i^k < U_j^k\} - \mathbf{1}\{U_j^k < U_i^k\}.$$

We write $Y^k = (Y_e^k)_{e \in \mathcal{E}}$, $\mathbf{Y} = (Y^1, \dots, Y^m) \in \{-1, 1\}^{\mathcal{E} \times [m]}$, and $\bar{Y} := \frac{1}{m} \sum_{k=1}^m Y^k$.

Given $\mathbf{Y}$, we consider partitioning the voters into two groups $\mathcal{V}_+$ and $\mathcal{V}_-$, such that $\mathcal{V}_+$ has strict majority order $1 \succ 2 \succ \dots \succ n$, while $\mathcal{V}_-$ has the reverse strict majority order. The partition can be equivalently characterized by a vector $s \in \{-1, +1\}^m$ and setting

$$\mathcal{V}_+ := \{k \in [m] : s_k = 1\}, \quad \mathcal{V}_- := \{k \in [m] : s_k = -1\}.$$

The pairwise margins for these groups are

$$M_{ij}^+ = \sum_{k \in \mathcal{V}_+} Y_{ij}^k = \frac{m \bar{Y}_{ij} + (\mathbf{Y}s)_{ij}}{2}, \quad M_{ij}^- = \sum_{k \in \mathcal{V}_-} Y_{ij}^k = \frac{m \bar{Y}_{ij} - (\mathbf{Y}s)_{ij}}{2}.$$

Therefore, if we can prove that there exists $s \in \{-1, 1\}^m$ such that for every $i < j$,

$$(13) \quad (\mathbf{Y}s)_{ij} \geq m |\bar{Y}_{ij}| + a_{ij} \quad a_{ij} > 0,$$

then $M_{ij}^+ > 0$ and $M_{ij}^- < 0$ for every $i < j$, and the two groups satisfied the desired order.

Finding such $s \in \{-1, 1\}^m$ is an integer programming problem, which is hard to solve directly. Therefore, we relax this problem to finding $\theta \in [-1, 1]^m$ first and then rounding θ to $\{-1, 1\}^m$. The relaxed problem is convex and can be easily characterized by the minimax theorem.

Lemma 3.1. *Let $a_e > 0$ for all $e \in \mathcal{E}$. Define the simplex $\Sigma_a^+ := \{w \in \mathbb{R}_+^{\mathcal{E}} : \langle a, w \rangle = 1\}$. Then there exists $\theta \in [-1, 1]^m$ satisfying*

$$(\mathbf{Y}\theta)_{ij} \geq m |\bar{Y}_{ij}| + \Gamma_a(\mathbf{Y}) a_{ij} \quad \forall i < j.$$

where

$$\Gamma_a(\mathbf{Y}) = \min_{w \in \Sigma_a^+} \left(\sum_{k=1}^m |\langle w, Y^k \rangle| - m \langle |\bar{Y}|, w \rangle \right).$$

Proof. Define

$$\Gamma_a(\mathbf{Y}) := \max_{\theta \in [-1, 1]^m} \min_{i < j} \frac{(\mathbf{Y}\theta)_{ij} - m |\bar{Y}_{ij}|}{a_{ij}}.$$

For fixed θ , the minimum over pairs equals the minimum of the same linear functional over the weighted simplex:

$$\min_{i < j} \frac{(\mathbf{Y}\theta)_{ij} - m |\bar{Y}_{ij}|}{a_{ij}} = \min_{w \in \Sigma_a^+} (\langle \mathbf{Y}\theta, w \rangle - m \langle |\bar{Y}|, w \rangle).$$

The sets $[-1, 1]^m$ and Σ_a^+ are compact and convex, and the objective is affine in each variable. The finite-dimensional minimax theorem gives

$$\begin{aligned}\Gamma_a(\mathbf{Y}) &= \min_{w \in \Sigma_a^+} \max_{\theta \in [-1, 1]^m} \left(\sum_{k=1}^m \theta_k \langle w, Y^k \rangle - m \langle |\bar{Y}|, w \rangle \right) \\ &= \min_{w \in \Sigma_a^+} \left(\sum_{k=1}^m |\langle w, Y^k \rangle| - m \langle |\bar{Y}|, w \rangle \right).\end{aligned}$$

□

The quantity $\Gamma_a(\mathbf{Y})$ characterizes the maximum amount of rounding error we could tolerate. To understand it, we need to understand the geometry of Y first.

Lemma 3.2. *For any edge vector $w \in \mathbb{R}^\mathcal{E}$, define the (weighted) norm*

$$\|w\| := \sqrt{\sum_{i < j} w_{ij}^2 + \sum_{i=1}^n (\mathbf{R}_i(w))^2},$$

with the divergence operator $\mathbf{R}_i(w)$ at vertex i given by $\mathbf{R}_i(w) := \sum_{j > i} w_{ij} - \sum_{j < i} w_{ji}$. Then for one voter, we have

$$\mathbb{E} \langle w, Y \rangle^2 = \frac{1}{3} \|w\|^2.$$

Moreover, we have

$$\mathbb{E} \langle w, Y \rangle^4 \leq 1584 (\mathbb{E} \langle w, Y \rangle^2)^2.$$

Define the non-negative unit sphere $\mathcal{S}_1^+ := \{w \in \mathbb{R}^\mathcal{E} : \|w\| = 1\}$ under this weighted norm. For an edge vector $z = (z_e)_{e \in \mathcal{E}} \in \mathbb{R}^\mathcal{E}$, define the (constrained) dual gauge $\|\cdot\|_{*,+}$ as the support function of the set $\mathcal{S}_1^+$:

$$\|z\|_{*,+} := \sup_{w \in \mathcal{S}_1^+} \langle z, w \rangle.$$

The following lemma controls the dual gauge of $1_\mathcal{E}$, $\bar{Y}$, and the Rademacher average of $\mathbf{Y}$:

Lemma 3.3. *The following inequalities hold:*

- (i) $\|1_\mathcal{E}\|_{*,+} \leq \sqrt{\frac{5}{6}} n^{5/6}$.
- (ii) *Let the random vector $G = (G_e)_{e \in \mathcal{E}}$ be defined by*

$$G_e := \sum_{k=1}^m \varepsilon_k Y_e^k,$$

where either $\varepsilon_k \equiv 1$ for all $k \in [m]$, or $\varepsilon_1, \dots, \varepsilon_m$ are Rademacher variables independent of each other and $\mathbf{Y}$. There exists c such that

$$\mathbb{E} \|G\|_{*,+}^2 \leq cmn^{5/3}.$$

In Section 3.3, we explain why the exponent $5/3$ should be expected for “reasonable” vectors (e.g. vectors with bounded coordinates) and provide the proof of Lemma 3.3. With these calculations, we can obtain the following high-probability events using standard concentration inequalities.

Theorem 3.2. *There exists c_0, c_1, c_2, c_3 such that for $\mathcal{G}_1, \mathcal{G}_2$ defined below*

$$\mathcal{G}_1 := \left\{ \frac{1}{m} \sum_{k=1}^m |\langle w, Y^k \rangle| \geq c_0 \quad \forall w \in \mathcal{S}_1^+ \right\}, \quad \mathcal{G}_2 := \left\{ \|\bar{Y}\|_{*,+} \leq \frac{c_0}{2} \right\},$$

whenever $m \geq c_1 n^{5/3}$, we have

$$\mathbb{P}(\mathcal{G}_1^c) \leq \exp(-c_2 m), \quad \mathbb{P}(\mathcal{G}_2^c) \leq \exp\left(-c_3 \frac{m}{n}\right).$$

The proof can be found in Section 3.1. Let $\mathcal{G} := \mathcal{G}_1 \cap \mathcal{G}_2$. Consequently, there exists c_4 , such that

$$\mathbb{P}(\mathcal{G}^c) \leq 2 \exp\left(-c_4 \frac{m}{n}\right).$$

On $\mathcal{G}_2$, the definition of $\|\cdot\|_{*,+}$ and homogeneity give that for every $w \in \mathbb{R}_+^\mathcal{E}$,

$$\langle |\bar{Y}|, w \rangle \leq \|\bar{Y}\|_{*,+} \|w\| \leq \frac{c_0}{2} \|w\|.$$

Subtracting this inequality from the defining inequality of $\mathcal{G}_1$ proves that, on $\mathcal{G}$,

$$(14) \quad \frac{1}{m} \sum_{k=1}^m |\langle w, Y^k \rangle| - \langle w, |\bar{Y}| \rangle \geq \frac{c_0}{2} \|w\| \quad \forall w \in \mathbb{R}_+^{\mathcal{E}}.$$

Combining (14) and Lemma 3.1, we obtain the following characterization of $\Gamma_a(\mathbf{Y})$: with high probability,

$$(15) \quad \Gamma_a(\mathbf{Y}) \gtrsim m \|a\|_{*,+}^{-1}.$$

Next, we quantify the rounding error in Theorem 3.3, whose proof is provided in Section 3.5.

Theorem 3.3. *Let $v_e \in [-1, 1]^m$ and $a_e \geq 1$ for all $e \in \mathcal{E}$. Suppose that for every $\gamma \geq 1$,*

$$(16) \quad \sum_{e \in \mathcal{E}} e^{-\gamma a_e^2} \leq m e^{-\gamma}.$$

Then there exists c such that for every $\theta \in [-1, 1]^m$, one can find $s \in \{-1, 1\}^m$ satisfying that for all $e \in \mathcal{E}$,

$$|\langle s - \theta, v_e \rangle| \leq c a_e \sqrt{m}.$$

Unfortunately, (16) is not satisfied by vectors with bounded coordinates. The following lemma provides a candidate and its proof is provided in Section 3.4.

Lemma 3.4. *Let $a := (a_{ij})_{i < j}$ where*

$$(17) \quad a_{ij} := \max \left\{ 1, \frac{2(j-i)}{n^{2/3}} \right\},$$

then we have

$$(18) \quad \|a\|_{*,+} \leq \sqrt{\frac{5}{6}} n^{5/6}.$$

Moreover, there exists c such that, for every $\gamma \geq 1$,

$$(19) \quad \sum_{i < j} e^{-\gamma a_{ij}^2} \leq c n^{5/3} e^{-\gamma}.$$

Remark 3.1. *It is worth noting that the choice in (17) is not arbitrary. (16) requires an edge weight vector $a \in \mathbb{R}_+^{\mathcal{E}}$ that grows monotonically with the spatial distance $(j-i)$. However, increasing edge weights increases the dual gauge $\|a\|_{*,+}$. If we choose a linear growth $a_{ij} \propto (j-i)/n^\gamma$, then the dual gauge will grow faster than $n^{5/6}$ for $\gamma < 2/3$. Consequently, (17) is the maximal allowable linear growth that safely preserves the expected $n^{5/6}$ dual gauge geometry.*

Now we can finish the proof of Theorem 3.1. Let $m \geq c_1 n^{5/3}$, where c_1 is the constant from Theorem 3.2. Choose a as in (17). Then (18) and (15) imply that there exists $\theta \in [-1, 1]^m$ such that

$$(\mathbf{Y}\theta)_{ij} \geq m |\bar{Y}_{ij}| + c_2 m n^{-5/6} a_{ij} \quad \text{for every } i < j.$$

For $e \in \mathcal{E}$, set

$$v_e := (Y_e^{(1)}, \dots, Y_e^{(m)}) \in [-1, 1]^m.$$

Increase c_1 if necessary. Provided $m \geq c_1 n^{5/3}$, (19) means that (16) holds for a . Theorem 3.3 therefore gives $s \in \{-1, 1\}^m$ such that, simultaneously for every $e = (i, j)$,

$$|(\mathbf{Y}(s - \theta))_{ij}| = |\langle s - \theta, v_e \rangle| \leq c_3 a_{ij} \sqrt{m}.$$

Hence,

$$(\mathbf{Y}s)_{ij} \geq m |\bar{Y}_{ij}| + a_{ij} (c_2 m n^{-5/6} - c_3 \sqrt{m}).$$

We may increase c_1 so that the quantity in parentheses is strictly positive. Thus (13) holds and the proof is completed.

3.1. Proof of Theorem 3.2. We are going to use the following concentration inequality in [Bou02, Theorem 2.3]. [\[zk: citation\]](#)

Lemma 3.5 (Bousquet's inequality). *Let $X_1, \dots, X_m$ be independent identically distributed random elements, and let $\mathcal{H}$ be a class of measurable functions such that, for every $h \in \mathcal{H}$, $\mathbb{E}h(X_1) = 0$, $|h(X_1)| \leq b$ almost surely, and $\mathbb{E}h(X_1)^2 \leq \sigma^2$. Define the random variable*

$$Z := \sup_{h \in \mathcal{H}} \sum_{k=1}^m h(X_k),$$

and suppose that Z is integrable. Then, for every $t > 0$,

$$\mathbb{P}(Z \geq \mathbb{E}Z + t) \leq \exp\left(-\frac{t^2}{2(m\sigma^2 + 2b\mathbb{E}Z + bt/3)}\right).$$

Proof of Theorem 3.2. The proof is separated into two parts.

Bound for $\mathcal{G}_1^c$. We are going to apply the bounded differences inequality to establish the concentration. The bounded differences of the absolute-value function would involve $\sup_{w \in \mathcal{S}_1^+} |\langle w, Y \rangle|$ which grows with n . Therefore, to obtain a dimension-free bounded-difference estimate, we first replace the absolute-value function by a bounded Lipschitz surrogate.

Fix $\kappa_0 > 0$, to be determined later, let $\phi : \mathbb{R} \rightarrow [0, 1]$ be the even piecewise-linear function

$$\phi(t) := \begin{cases} 0, & |t| \leq \kappa_0/2 \\ 2|t|/\kappa_0 - 1, & \kappa_0/2 < |t| < \kappa_0 \\ 1, & |t| \geq \kappa_0 \end{cases}.$$

It satisfies $\phi(0) = 0$, $\text{Lip}(\phi) = 2/\kappa_0$, and $|t| \geq \frac{\kappa_0}{2}\phi(t)$. Therefore, we have

$$(20) \quad \frac{1}{m} \sum_{k=1}^m |\langle w, Y^k \rangle| \geq \frac{\kappa_0}{2m} \sum_{k=1}^m \phi(\langle w, Y^k \rangle).$$

It remains to provide a lower bound on $\frac{1}{m} \sum_{k=1}^m \phi(\langle w, Y^k \rangle)$ with high probability. We first prove that there exists universal constants $\kappa_0, p_0 > 0$, such that for any $w \in \mathcal{S}_1^+$,

$$(21) \quad \mathbb{P}(|\langle w, Y \rangle| \geq \kappa_0) \geq p_0.$$

Indeed, since $\|w\| = 1$, Lemma 3.2 gives $\mathbb{E}\langle w, Y \rangle^2 = 1/3$. Thus, applying Paley–Zygmund inequality to $\langle w, Y \rangle^2$ with parameter $1/4$ and using Lemma 3.2 gives

$$\mathbb{P}\left(\langle w, Y \rangle^2 \geq \frac{1}{12}\right) = \mathbb{P}\left(\langle w, Y \rangle^2 \geq \frac{1}{4}\mathbb{E}\langle w, Y \rangle^2\right) \geq \left(1 - \frac{1}{4}\right)^2 \frac{(\mathbb{E}\langle w, Y \rangle^2)^2}{\mathbb{E}\langle w, Y \rangle^4} \geq \frac{9}{16 \cdot 1584} := p_0.$$

This implies (21). By the definition of $\phi(\cdot)$, (21) gives $\mathbb{E}\phi(\langle w, Y \rangle) \geq p_0$. Define

$$\Delta := \sup_{w \in \mathcal{S}_1^+} \left| \frac{1}{m} \sum_{k=1}^m (\phi(\langle w, Y^k \rangle) - \mathbb{E}\phi(\langle w, Y \rangle)) \right|.$$

It remains to provide an upper bound on Δ with high probability. We first give an upper bound on $\mathbb{E}\Delta$. Let $\varepsilon_1, \dots, \varepsilon_m$ be independent Rademacher variables. Applying symmetrization and the contraction principle gives

$$\mathbb{E}\Delta \leq \frac{8}{\kappa_0 m} \mathbb{E} \sup_{w \in \mathcal{S}_1^+} \left| \sum_{k=1}^m \varepsilon_k \langle w, Y^k \rangle \right| \leq \frac{8}{\kappa_0 m} \mathbb{E} \left\| \sum_{k=1}^m \varepsilon_k Y^k \right\|_{*,+} \leq c_1 \sqrt{\frac{n^{5/3}}{m}},$$

where the last line uses Lemma 3.3 and Jensen's inequality. Choosing c_2 large enough that $m \geq c_2 n^{5/3}$ implies $\mathbb{E}\Delta \leq p_0/4$.

Regard Δ as a function of the m independent random variables $Y^1, \dots, Y^m$. If one of $Y^1, \dots, Y^m$ is replaced, only one summand in the empirical average $\frac{1}{m} \sum_{k=1}^m \phi(\langle w, Y^k \rangle)$ changes. Since $0 \leq \phi \leq 1$, this average changes by at most $1/m$. Taking the absolute value and the supremum over w does not increase this bound. Since $\mathbb{E}\Delta \leq p_0/4$, applying bounded difference inequality gives

$$\mathbb{P}\left(\Delta > \frac{p_0}{2}\right) \leq \mathbb{P}\left(\Delta - \mathbb{E}\Delta > \frac{p_0}{4}\right) \leq \exp\left(-\frac{2(p_0/4)^2}{\sum_{k=1}^m (1/m)^2}\right) = e^{-c_3 m}.$$

On the complementary event, we have

$$\frac{1}{m} \sum_{k=1}^m \phi(\langle w, Y^k \rangle) \geq \mathbb{E}[\phi(\langle w, Y \rangle)] - \Delta \geq p_0 - \frac{p_0}{2} = \frac{p_0}{2}.$$

Plugging this into (20) gives the desired result.

Bound for $\mathcal{G}_2^c$. Notice that Lemma 3.3 and Jensen's inequality give $\mathbb{E} \|\bar{Y}\|_{*,+} \leq c_1 \sqrt{\frac{n^{5/3}}{m}}$. Therefore, if $m \geq c_2 n^{5/3}$, then $\mathbb{E} \|\bar{Y}\|_{*,+} \leq \frac{c_0}{4}$. Hence,

$$(22) \quad \mathbb{P}(\mathcal{G}_2^c) = \mathbb{P}\left(\|\bar{Y}\|_{*,+} > \frac{c_0}{2}\right) \leq \mathbb{P}\left(Z - \mathbb{E}Z \geq \frac{c_0 m}{4}\right),$$

where $Z := m \|\bar{Y}\|_{*,+}$. We are going to apply Lemma 3.5 to Z . Note that

$$Z = \sup_{w \in S_1^+} \sum_{e \in \mathcal{E}} w_e \left| \sum_{k=1}^m Y_e^k \right| = \sup_{\substack{w \in S_1^+ \\ \varepsilon \in \{-1,1\}^\mathcal{E}}} \sum_{e \in \mathcal{E}} w_e \varepsilon_e \left(\sum_{k=1}^m Y_e^k \right) = \sup_{\substack{w \in S_1^+ \\ \varepsilon \in \{-1,1\}^\mathcal{E}}} \sum_{k=1}^m \langle \varepsilon \odot w, Y^k \rangle.$$

For simplicity, let $h_{w,\varepsilon}(Y) := \langle \varepsilon \odot w, Y \rangle$, where $\odot$ denotes the coordinate-wise product. Since $\mathbb{E}Y^k = 0$, it follows that $\mathbb{E}h_{w,\varepsilon}(Y^k) = 0$. By Lemma 3.3, we have

$$|h_{w,\varepsilon}(Y)| \leq \sum_{e \in \mathcal{E}} w_e \leq \|1_\mathcal{E}\|_{*,+} \leq n^{5/6}.$$

Moreover, by Cauchy-Schwarz,

$$\begin{aligned} \mathbb{E}[(h_{w,\varepsilon}(Y))^2] &= \frac{\|\varepsilon \odot w\|^2}{3} \leq \frac{1}{3} \left(\sum_{i < j} w_{ij}^2 + \sum_{i=1}^n \left(\sum_{j > i} w_{ij} + \sum_{j < i} w_{ji} \right)^2 \right) \\ &\leq \frac{2n-1}{3} \sum_{i < j} w_{ij}^2 \leq n. \end{aligned}$$

Therefore Lemma 3.5 gives

$$(23) \quad \mathbb{P}\left(Z - \mathbb{E}Z \geq \frac{c_0 m}{4}\right) \leq \exp\left(-\frac{c_0^2 m^2}{64(mn + c_0 m + c_0 mn^{5/6})}\right) \leq \exp\left(-\frac{c_0^2 m}{64(2c_0 + 1)n}\right).$$

Plugging (23) into (22) proves the desired result. $\square$

3.2. Proof of Lemma 3.2. Fix $w \in \mathbb{R}^\mathcal{E}$ and put $A_w := \sum_{i < j} w_{ij}^2$ and $B_w := \sum_{i=1}^n \mathbf{R}_i(w)^2$, so that $\|w\|^2 = A_w + B_w$. The decomposition (2) gives

$$Y_{ij} = g(U_i) - g(U_j) + H_{ij},$$

where

$$H_{ij} = \begin{cases} 1 - 2(U_j - U_i), & U_i < U_j, \\ -1 + 2(U_i - U_j), & U_i > U_j, \end{cases}$$

so $|H_{ij}| \leq 1$.

For fixed $u \in [0, 1]$, we have $\mathbb{E}[Y_{ij} | U_i = u] = g(u)$ and $\mathbb{E}[Y_{ij} | U_j = u] = -g(u)$. Since $\mathbb{E}g(U_i) = 0$, it follows that

$$(24) \quad \mathbb{E}[H_{ij} | U_i] = \mathbb{E}[H_{ij} | U_j] = 0.$$

Collecting the coefficient of each $g(U_i)$ in $\langle w, Y \rangle$ gives the Hoeffding decomposition

$$\langle w, Y \rangle = L_w + D_w,$$

where $L_w := \sum_{i=1}^n \mathbf{R}_i(w)g(U_i)$ and $D_w := \sum_{i < j} w_{ij}H_{ij}$.

We first compute the second moment. The random variables $g(U_i)$ are independent and centered, with $\mathbb{E}g(U_i)^2 = 1/3$. By (24), the three terms in $Y_{ij} = g(U_i) - g(U_j) + H_{ij}$ are pairwise orthogonal. Since $\mathbb{E}Y_{ij}^2 = 1$, this gives $\mathbb{E}H_{ij}^2 = 1/3$. By Lemma 2.3, if two distinct edges are disjoint, their H -variables are independent and centered; if they share one endpoint, they are orthogonal. Consequently, $\mathbb{E}L_w^2 = B_w/3$ and $\mathbb{E}D_w^2 = A_w/3$. As every $g(U_i)$ is orthogonal to every $H_{h\ell}$, we have $\mathbb{E}[L_w D_w] = 0$. Therefore,

$$\mathbb{E}\langle w, Y \rangle^2 = \frac{A_w + B_w}{3} = \frac{\|w\|^2}{3}.$$

We next prove the fourth-moment estimate. Since $g(U)$ is uniform on $[-1, 1]$, $\mathbb{E}g(U)^2 = 1/3$ and $\mathbb{E}g(U)^4 = 1/5$. Expanding the fourth power and using independence gives

$$\begin{aligned} \mathbb{E}L_w^4 &= \frac{1}{5} \sum_{i=1}^n (\mathbf{R}_i(w))^4 + \frac{2}{3} \sum_{i < j} (\mathbf{R}_i(w))^2 (\mathbf{R}_j(w))^2 \\ (25) \quad &\leq \frac{1}{3} \left(\sum_{i=1}^n (\mathbf{R}_i(w))^2 \right)^2 = \frac{B_w^2}{3}. \end{aligned}$$

To bound $\mathbb{E}D_w^4$, let $q = (e_1, e_2, e_3, e_4) \in \mathcal{E}^4$ be a tuple of four edges, and $\Theta(q) := \prod_{\ell=1}^4 w_{e_\ell}$ and $M(q) := \mathbb{E} \prod_{\ell=1}^4 H_{e_\ell}$ be its corresponding weight and expectation respectively. Then,

$$(26) \quad \mathbb{E}D_w^4 = \sum_{q \in \mathcal{E}^4} \Theta(q) M(q).$$

Given q , we can define a multigraph on $[n]$ by including the four edges in q . For an edge $f \in \mathcal{E}$, let $\mu_q(f) := |\{\ell \in [4] : e_\ell = f\}|$ be its multiplicity. For a vertex $v \in [n]$, its degree is $d_q(v) := \sum_{i < v} \mu_q((i, v)) + \sum_{j > v} \mu_q((v, j))$. If $d_q(v) = 1$, there is a unique $\ell_0 \in [4]$ and a unique neighbor u such that e_{ℓ_0} joins u and v . Conditioning first on $(U_h)_{h \neq v}$ and then using (24) gives $M(q) = 0$.

For distinct vertices $u, v \in [n]$, let $e(u, v) \in \mathcal{E}$ be the edge connecting them. Define the families of triangle and four-cycle edge sets by

$$\begin{aligned} \mathfrak{C}_3 &:= \{ \{e(a, b), e(b, c), e(c, a)\} : a, b, c \text{ are distinct} \}, \\ \mathfrak{C}_4 &:= \{ \{e(a, b), e(b, c), e(c, d), e(d, a)\} : a, b, c, d \text{ are distinct} \}. \end{aligned}$$

Let $\rho(q)$ be the nonincreasing list of the positive multiplicities $\mu_q(f)$. Since the total multiplicity is four, the absence of a degree-one vertex leaves exactly the following possibilities:

$$M(q) \neq 0 \implies \begin{cases} \rho(q) = (4), & \text{type 1,} \\ \rho(q) = (2, 2), & \text{type 2,} \\ \rho(q) = (2, 1, 1), \text{ supp}(q) \in \mathfrak{C}_3, & \text{type 3,} \\ \rho(q) = (1, 1, 1, 1), \text{ supp}(q) \in \mathfrak{C}_4, & \text{type 4.} \end{cases}$$

Indeed, every nonisolated vertex has degree at least two. A support with one or two edges gives the first two types; a three-edge support must be a triangle with one repeated edge; and a four-edge support must be a simple four-cycle.

Fix a total order on $\mathcal{E}$, and let τ_ℓ be the sum of the absolute values of the terms of type ℓ in (26). Since $|H_e| \leq 1$,

$$\tau_1 \leq \sum_{e \in \mathcal{E}} w_e^4 \leq A_w^2, \quad \tau_2 \leq 6 \sum_{e < f} w_e^2 w_f^2 \leq 3A_w^2.$$

For type 3, the inequality $a^2bc + b^2ac + c^2ab \leq a^2b^2 + a^2c^2 + b^2c^2$ for nonnegative a, b, c gives

$$\begin{aligned} \tau_3 &\leq 12 \sum_{\{e, f, h\} \in \mathfrak{C}_3} (w_e^2 |w_f| |w_h| + w_f^2 |w_e| |w_h| + w_h^2 |w_e| |w_f|) \\ &\leq 12 \sum_{\{e, f, h\} \in \mathfrak{C}_3} (w_e^2 w_f^2 + w_e^2 w_h^2 + w_f^2 w_h^2). \end{aligned}$$

Every pair of distinct edges sharing a vertex belongs to a unique triangle, and therefore

$$\sum_{\{e, f, h\} \in \mathfrak{C}_3} (w_e^2 w_f^2 + w_e^2 w_h^2 + w_f^2 w_h^2) \leq \sum_{e < f} w_e^2 w_f^2 \leq \frac{A_w^2}{2}.$$

Thus $\tau_3 \leq 6A_w^2$.

For type 4, let $\mathbf{C}_w \in \mathbb{R}^{n \times n}$ be the symmetric matrix with $(\mathbf{C}_w)_{ij} := |w_{e(i, j)}|$ for $i \neq j$ and zero diagonal. Then $\text{tr}(\mathbf{C}_w^2) = 2A_w$. Since $\mathbf{C}_w$ is real symmetric, there are an orthogonal matrix $\mathbf{U}$ and real numbers $\lambda_1, \dots, \lambda_n$ such that $\mathbf{C}_w = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top$. Consequently,

$$\text{tr}(\mathbf{C}_w^4) = \sum_{i=1}^n \lambda_i^4 \leq \left(\sum_{i=1}^n \lambda_i^2 \right)^2 = (\text{tr}(\mathbf{C}_w^2))^2 = 4A_w^2.$$

Each member of $\mathfrak{C}_4$ occurs exactly eight times in the trace expansion, corresponding to four starting vertices and two orientations. All terms in that expansion are nonnegative, so

$$8 \sum_{C \in \mathfrak{C}_4} \prod_{e \in C} |w_e| \leq \text{tr}(\mathbf{C}_w^4) \leq 4A_w^2.$$

There are $4! = 24$ orderings of the four distinct edges in (26); hence

$$\tau_4 \leq 24 \sum_{C \in \mathfrak{C}_4} \prod_{e \in C} |w_e| \leq 12A_w^2.$$

Combining the four types gives

$$(27) \quad \mathbb{E}D_w^4 \leq \tau_1 + \tau_2 + \tau_3 + \tau_4 \leq 22A_w^2.$$

Finally, using $(a+b)^4 \leq 8(a^4 + b^4)$ together with (25) and (27), we obtain

$$\begin{aligned} \mathbb{E}\langle w, Y \rangle^4 &\leq 8(\mathbb{E}L_w^4 + \mathbb{E}D_w^4) \\ &\leq 8 \left(\frac{B_w^2}{3} + 22A_w^2 \right) \leq 176(A_w + B_w)^2 = 1584(\mathbb{E}\langle w, Y \rangle^2)^2. \end{aligned}$$

This completes the proof.

3.3. Proof of Lemma 3.3. We first state a dual representation of the dual gauge.

Lemma 3.6. *For any $z \in \mathbb{R}_+^\mathcal{E}$, the dual gauge satisfies:*

$$(28) \quad \|z\|_{*,+}^2 = \min_{u \in \mathbb{R}^n} \left(\sum_{i=1}^n u_i^2 + \sum_{i < j} (z_{ij} - u_i + u_j)_+^2 \right).$$

Proof. Since $z \in \mathbb{R}_+^\mathcal{E}$, we have $\|z\|_{*,+} \geq 0$. Therefore, evaluating the squared gauge is equivalent to optimizing the convex conjugate:

$$\frac{1}{2} \|z\|_{*,+}^2 = \max_{w \in \mathbb{R}_+^\mathcal{E}} \left\{ \langle z, w \rangle - \frac{1}{2} \|w\|^2 \right\}.$$

Using the definition of $\|w\|$ and introducing $v_i = R_i(w)$, the right-hand side equals

$$\max_{\substack{w \in \mathbb{R}_+^\mathcal{E}, v \in \mathbb{R}^n \\ v_i = R_i(w), i \in [n]}} \left\{ \sum_{i < j} z_{ij} w_{ij} - \frac{1}{2} \sum_{i < j} w_{ij}^2 - \frac{1}{2} \sum_{i=1}^n v_i^2 \right\}.$$

Next, we introduce Lagrange multipliers $u \in \mathbb{R}^n$ to relax the equality constraints. The corresponding convex minimization problem satisfies Slater's condition: for example, one may take $w_{ij} = 1$ for every $i < j$ and $v_i = R_i(w)$. Hence strong Lagrangian duality applies. Using $\sum_{i=1}^n u_i R_i(w) = \sum_{i < j} (u_i - u_j) w_{ij}$, we obtain

$$\begin{aligned} \frac{1}{2} \|z\|_{*,+}^2 &= \inf_{u \in \mathbb{R}^n} \left[\sum_{i=1}^n \max_{v_i \in \mathbb{R}} \left(u_i v_i - \frac{1}{2} v_i^2 \right) + \sum_{i < j} \max_{w_{ij} \geq 0} \left((z_{ij} - u_i + u_j) w_{ij} - \frac{1}{2} w_{ij}^2 \right) \right] \\ &= \frac{1}{2} \inf_{u \in \mathbb{R}^n} \left(\sum_{i=1}^n u_i^2 + \sum_{i < j} (z_{ij} - u_i + u_j)_+^2 \right), \end{aligned}$$

The final objective is continuous and is bounded below by $\sum_{i=1}^n u_i^2$, so it is coercive and its infimum is attained. This gives the desired result. $\square$

Using (28), we can prove Lemma 3.3 by choosing the optimal $u \in \mathbb{R}^n$. However, the true minimizer is hard to compute. We instead seek a rate optimal ansatz when n is large. Take $1_\mathcal{E}$ as an example, we have

$$\|1_\mathcal{E}\|_{*,+}^2 = \min_{u \in \mathbb{R}^n} J(u), \quad J(u) := \sum_{i=1}^n u_i^2 + \sum_{i < j} (1 - u_i + u_j)_+^2.$$

Heuristically, assume that $u_i = \lambda f(\frac{i}{n})$ for a smooth function $f : [0, 1] \rightarrow \mathbb{R}$. Then, as $n \rightarrow \infty$,

$$J(f) \approx n\lambda^2 \int_0^1 f(x)^2 dx + n^2 \int_0^1 \int_0^1 (1 + \lambda f'(x)y)_+^2 dy dx$$

$$\approx n\lambda^2 \int_0^1 f(x)^2 dx + \frac{n^2}{3\lambda} \int_0^1 \frac{-1}{f'(x)} dx.$$

By choosing the optimal scaling $\lambda \asymp n^{1/3}$, we have

$$J(f) \asymp n^{5/3} \left(\int_0^1 f(x)^2 dx \right)^{1/3} \left(\int_0^1 \frac{-1}{f'(x)} dx \right)^{2/3},$$

obtaining the correct exponent $5/3$.

Proof of Lemma 3.3. For 1_ε , choose $u_i = (n+1-2i)/n^{2/3}$. Then $u_i - u_j = 2(j-i)/n^{2/3}$ and $\sum_i u_i^2 = n(n^2-1)/(3n^{4/3}) \leq n^{5/3}/3$. Applying (28) to $z = 1_\varepsilon$ yields

$$\begin{aligned} \|1_\varepsilon\|_{*,+}^2 &\leq \frac{n^{5/3}}{3} + \sum_{i < j} \left(1 - \frac{2(j-i)}{n^{2/3}} \right)_+ \\ &= \frac{n^{5/3}}{3} + \sum_{k=1}^{n-1} (n-k) \left(1 - \frac{2k}{n^{2/3}} \right)_+^2 \\ &\leq \frac{n^{5/3}}{3} + \frac{n^{5/3}}{2} = \frac{5}{6} n^{5/3}. \end{aligned}$$

For G , note that G_{ij} is a sum of m independent Rademacher variables for each fixed $i < j$. In the case of random ε_k , this statement holds conditionally on the rankings. Thus $\mathbb{P}(|G_{ij}| \geq t) \leq 2e^{-t^2/(2m)}$. For every $b \geq 0$, tail integration gives

$$(29) \quad \begin{aligned} \mathbb{E}(|G_{ij}| - b)_+^2 &= 2 \int_b^\infty (t-b) \mathbb{P}(|G_{ij}| \geq t) dt \\ &\leq 4me^{-b^2/(2m)}. \end{aligned}$$

Choose $u_i := \sqrt{m}(n+1-2i)/n^{2/3}$. Then $u_i - u_j = 2\sqrt{m}(j-i)/n^{2/3}$ and $\sum_i u_i^2 \leq mn^{5/3}/3$. Applying (28) to $z = |G|$ and using (29) gives

$$\begin{aligned} \mathbb{E} \| |G| \|_{*,+}^2 &\leq \sum_i u_i^2 + 4m \sum_{i < j} e^{-2(j-i)^2/n^{4/3}} \\ &\leq \frac{1}{3} mn^{5/3} + 4cmn^{5/3} \leq c' mn^{5/3}. \end{aligned}$$

□

3.4. Proof of Lemma 3.4. Define $L := n^{2/3}$ and $u \in \mathbb{R}^n$ by $u_i := (n+1-2i)/L$. For $i < j$, $u_i - u_j = 2(j-i)/L$. Hence, with $\psi_{ij} := (1 - u_i + u_j)_+$, we have $a_{ij} = u_i - u_j + \psi_{ij}$, and therefore, by (28),

$$\|a\|_{*,+}^2 \leq \sum_i u_i^2 + \sum_{i < j} \psi_{ij}^2.$$

Now $\sum_i u_i^2 = n(n^2-1)/(3L^2) \leq n^{5/3}/3$. Moreover, $\psi_{ij} > 0$ only when $j-i < L/2$, and $0 \leq \psi_{ij} \leq 1$, so $\sum_{i < j} \psi_{ij}^2 \leq nL/2 = n^{5/3}/2$.

For (19), write $k = j-i$ and use $n-k \leq n$:

$$\begin{aligned} \sum_{i < j} e^{-\gamma a_{ij}^2} &\leq n \sum_{k=1}^{n-1} \exp \left(-\gamma \max \left\{ 1, \frac{4k^2}{L^2} \right\} \right) \\ &\leq n \sum_{k=1}^{\infty} \exp \left(-\gamma \max \left\{ 1, \frac{4k^2}{L^2} \right\} \right). \end{aligned}$$

The terms with $k \leq L/2$ contribute at most $Le^{-\gamma}$. For $k > L/2$, the inequality $(2k/L)^2 \geq 2k/L$ gives $e^{-4\gamma k^2/L^2} \leq e^{-2\gamma k/L}$. Set $k_0 := \lfloor L/2 \rfloor + 1$ and $p := 2\gamma/L$. Since $pk_0 > \gamma$, the geometric-series bound gives

$$\sum_{k > L/2} e^{-4\gamma k^2/L^2} \leq \frac{e^{-\gamma}}{1 - e^{-p}}.$$

If $p \leq 1$, then $1 - e^{-p} \geq p/2$, so the last expression is at most $Le^{-\gamma}/\gamma \leq Le^{-\gamma}$ as $\gamma \geq 1$. If $p > 1$, it is at most $e^{-\gamma}/(1 - e^{-1}) \leq Le^{-\gamma}/(1 - e^{-1})$. Thus the sum over k is at most $cLe^{-\gamma}$ for a universal constant c . Multiplying by n and using $nL = n^{5/3}$ proves (19).

3.5. Proof of Theorem 3.3. We first restate the algorithmic theorem of Lovett and Meka in the form in which it appears in [LM15, Theorem 4].

Theorem 3.4 (Theorem 4 in [LM15]). *Let $v_e \in \mathbb{R}^m$ for all $e \in \mathcal{E}$, let $x_0 \in [-1, 1]^m$, and let $\lambda \in \mathbb{R}_+^\mathcal{E}$ such that*

$$\sum_{e \in \mathcal{E}} e^{-\lambda_e^2/16} \leq \frac{m}{16}.$$

For every approximation parameter $\delta > 0$, there exists a vector $x \in [-1, 1]^m$ satisfying

- (1) $|\langle x - x_0, v_e \rangle| \leq \lambda_e \|v_e\|_2$ for all $e \in \mathcal{E}$.
- (2) $|\{i \in [m] : |x_i| \geq 1 - \delta\}| \geq \frac{m}{2}$.

Through a standard compactness argument, we can get rid of the approximation parameter δ and obtain the following non-algorithmic consequence for the rounding argument.

Corollary 3.1. *Let $v_e \in \mathbb{R}^m$ for all $e \in \mathcal{E}$, let $x_0 \in [-1, 1]^m$, and let $\lambda \in \mathbb{R}_+^\mathcal{E}$ such that*

$$\sum_{e \in \mathcal{E}} e^{-\lambda_e^2/16} \leq \frac{m}{16}.$$

There exists $x \in [-1, 1]^m$ such that $|\{i \in [m] : |x_i| = 1\}| \geq \frac{m}{2}$, and $|\langle x - x_0, v_e \rangle| \leq \lambda_e \|v_e\|_2$ for all $e \in \mathcal{E}$.

Proof. Let $r := \lceil m/2 \rceil$, and let $\delta_k := 1/k$ for all $k \geq 1$. For each k , applying Theorem 3.4 with $\delta = \delta_k$ gives $x^{(k)} \in [-1, 1]^m$ such that $|\{i \in [m] : |x_i^{(k)}| \geq 1 - \delta_k\}| \geq r$, and for all $e \in \mathcal{E}$,

$$|\langle x^{(k)} - x_0, v_e \rangle| \leq \lambda_e \|v_e\|_2.$$

For each $k \geq 1$, choose a set $\mathcal{I}_k \subseteq [m]$ with $|\mathcal{I}_k| = r$ such that $|x_i^{(k)}| \geq 1 - \delta_k$ for all $i \in \mathcal{I}_k$. Since $[-1, 1]^m$ is compact, the sequence $(x^{(k)})_{k \geq 1}$ has a convergent subsequence. As there are only finitely many subsets of $[m]$ of cardinality r , after passing to a further subsequence we may assume that $x^{(k_\ell)} \rightarrow x$ and $\mathcal{I}_{k_\ell} = \mathcal{I}$ for some fixed $\mathcal{I} \subseteq [m]$ with $|\mathcal{I}| = r$. For every $i \in \mathcal{I}$, it follows that

$$|x_i| = \lim_{\ell \rightarrow \infty} |x_i^{(k_\ell)}| \geq \lim_{\ell \rightarrow \infty} (1 - \delta_{k_\ell}) = 1.$$

Since $x \in [-1, 1]^m$, we have $|x_i| = 1$ for all $i \in \mathcal{I}$. Thus,

$$|\{i \in [m] : |x_i| = 1\}| \geq |\mathcal{I}| = r \geq \frac{m}{2}.$$

Moreover, for every $e \in \mathcal{E}$, we have

$$|\langle x - x_0, v_e \rangle| = \lim_{\ell \rightarrow \infty} |\langle x^{(k_\ell)} - x_0, v_e \rangle| \leq \lambda_e \|v_e\|_2.$$

This completes the proof. $\square$

Now the stronger condition in (16) allows us to prove Theorem 3.3 by applying Corollary 3.1 iteratively.

Proof of Theorem 3.3. We use an iteration argument to prove this conclusion. Set $x^{(0)} := \theta$. At each stage $t \in \mathbb{N}_{\geq 0}$, let $I_t := \{r \in [m] : |x_r^{(t)}| < 1\}$ be the coordinates not yet fixed at ± 1 at this stage, and let $k_t := |I_t|$. Then $k_0 \leq m$. If $k_t = 0$, the iteration stops. Otherwise, for each $e \in \mathcal{E}$, restrict v_e to the active coordinates and write $u_e^{(t)} := (v_{e,r})_{r \in I_t} \in \mathbb{R}^{k_t}$. Define $\lambda_{e,t} := 4D_t a_e$ for all $e \in \mathcal{E}$ where

$$D_t := \sqrt{\log \left(\frac{16m}{k_t} \right)}.$$

Since $D_t \geq 1$. By (16), it follows that

$$\sum_{e \in \mathcal{E}} e^{-\lambda_{e,t}^2/16} = \sum_{e \in \mathcal{E}} e^{-D_t^2 a_e^2} \leq m e^{-D_t^2} = \frac{k_t}{16}.$$

Applying Corollary 3.1 on initial point $(x_r^{(t)})_{r \in I_t}$ and vectors $\{u_e^{(t)}\}_{e \in \mathcal{E}}$ with parameters $\lambda_{e,t}$, and then extending the resulting vector to $[m]$ by leaving all coordinates outside I_t unchanged, call the extension resulting vector $x^{(t+1)}$, then

$$(30) \quad k_{t+1} \leq \frac{k_t}{2}.$$

Moreover, for every $e \in \mathcal{E}$,

$$(31) \quad |\langle x^{(t+1)} - x^{(t)}, v_e \rangle| \leq \lambda_{e,t} \|u_e^{(t)}\|_2 \leq 4D_t a_e \sqrt{k_t}.$$

Due to (30), the iteration stops after finitely many stages. Denote the final vector by $s \in \{-1, 1\}^m$.

It now remains to sum (31). Group the values of k_t into the dyadic intervals

$$\frac{m}{2^{q+1}} < k_t \leq \frac{m}{2^q}, \quad q = 0, 1, 2, \dots$$

By (30), each interval contains at most one stage. If k_t lies in the q -th interval, then

$$D_t \sqrt{k_t} = \sqrt{\log \left(\frac{16m}{k_t} \right)} \cdot \sqrt{k_t} \leq \sqrt{(q+5) \log 2} \cdot \sqrt{\frac{m}{2^q}} = \sqrt{\frac{(q+5) \log 2}{2^q}} \cdot \sqrt{m}.$$

Consequently, for all $e \in \mathcal{E}$,

$$\begin{aligned} | \langle s - \theta, v_e \rangle | &\leq \sum_{t: k_t > 0} \left| \langle x^{(t+1)} - x^{(t)}, v_e \rangle \right| \leq 4a_e \sum_{t: k_t > 0} D_t \sqrt{k_t} \\ &\leq 4 \left(\sum_{q \geq 0} \sqrt{\frac{(q+5) \log 2}{2^q}} \right) a_e \sqrt{m} \leq c_1 a_e \sqrt{m}, \end{aligned}$$

for some universal constant $c_1 > 0$. This completes our proof. $\square$

APPENDIX A. PROOF OF LEMMA 2.4

Proof of Lemma 2.4. It is easy to see that $\mathbb{E}V_k^2 = 1/3$ and $\mathbb{E}V_k^4 = 1/5$. Independence and centering give

$$\begin{aligned} \mathbb{E}Y_s^2 &= \frac{1}{3}, \\ \mathbb{E}Y_s^4 &= \frac{1}{s^2} \left(s \cdot \frac{1}{5} + 3s(s-1) \cdot \frac{1}{9} \right) = \frac{1}{3} - \frac{2}{15s} \leq \frac{1}{3}. \end{aligned}$$

Applying the Paley–Zygmund inequality to Y_s^2 , we obtain

$$\mathbb{P} \left(|Y_s| \geq \frac{1}{4} \right) = \mathbb{P} \left(|Y_s|^2 \geq \frac{3}{16} \mathbb{E}Y_s^2 \right) \geq \left(1 - \frac{3}{16} \right)^2 \frac{(\mathbb{E}Y_s^2)^2}{\mathbb{E}Y_s^4} = \frac{169}{768}.$$

The law of Y_s is symmetric, so

$$\mathbb{P}(Y_s \leq -1/4) \geq p_0 := \frac{169}{1536}.$$

On this event $f(Y_s) = e^{-1}$, while everywhere $f \leq 1$. Therefore $\mathbb{E}f(Y_s) \leq 1 - (1 - e^{-1})p_0 =: \rho_0 < 1$. $\square$

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